

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Are these pairs of vectors orthogonal to each other or only independent?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 \neq 0 \quad \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 2 \neq 0$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 \neq 0 \Rightarrow \text{linearly independent only}$$

(b) $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix}.$

$$\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0.4 \\ 0.3 \end{bmatrix} = 0 \Rightarrow \text{orthogonal.}$$

(c) $\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix}.$

$$\begin{bmatrix} \frac{1}{2} \cos \theta \\ -\frac{1}{2} \sin \theta \\ \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} \sin \theta \\ \cos \theta \\ 0 \end{bmatrix} = \frac{1}{2} \cos \theta \sin \theta - \frac{1}{2} \sin \theta \cos \theta + 0 = 0$$

$$\Rightarrow \text{orthogonal}$$

Problem 2. (4 points) Given two vectors $\mathbf{a}_1 = (1, 1, 0)$ and $\mathbf{a}_2 = (1, 1, 1)$. Find the projection matrix $\mathbf{P}$ for the subspace of $\mathbb{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

Solution: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\Rightarrow (A^T A)^{-1} = \frac{1}{6-4} \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \\ = \begin{bmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{bmatrix}$$

$$\mathbf{P} = A(A^T A)^{-1} A^T \\ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$